

**Question 1 15 marks**

a) For  $f(x) = x^2 + 1 + \frac{1}{x^2}$  show that  $f(y) = f\left(\frac{1}{y}\right)$  2

b) State the natural domain and range of  $y = \sqrt{x-3}$  2

c) For the function:

$$f(x) = \begin{cases} 4, & x < -2 \\ x^2, & -2 \leq x \leq 2 \\ x-5, & x > 2 \end{cases}$$

(i) Sketch the graph of this function 2

(ii) Evaluate  $f(-5) + 3f(-2) - f(2) + 2f(4)$ . 2

d) For the function  $y = \frac{2}{x^2 - 9}$

(i) Determine whether the function is odd, even or neither. 2

(ii) Find the equation of the vertical asymptotes. 1

(iii) Find the equation of the horizontal asymptote. 1

(iv) Find the coordinates of any intercepts for this curve. 1

(v) Sketch the curve, showing all the above information. 2

*Question 2 on page 2.*

**Question 2** 15 marks Start this question on a new page.

- a) Solve for x:  $\frac{2x+3}{5x-1} = \frac{2x+1}{5x-5}$  3
- b) Solve algebraically the following simultaneously equations.  $x + 2y = -8$  and  $xy = 8$  3
- c) Solve for x:  $4 + 3x - x^2 > 0$ . 2
- c) (i) On the same diagram sketch  $y = |2x - 5|$  and  $y = x + 2$  2
- (ii) For what values of x is  $|2x - 5| \leq x + 2$  2
- d) Solve the following inequation. 3
- $$\frac{2}{x-3} < 4, x \neq 3$$

**Question 3** 15 marks Start this question on a new page.

- a) Show that  $\sin 45^\circ \cdot \cos 30^\circ + \cos 45^\circ \cdot \sin 30^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}$  3
- b) Solve for  $0^\circ \leq x \leq 360^\circ$ ,  $2 \sin x + \sqrt{3} = 0$  3
- c) If  $\tan \theta = \frac{7}{24}$  and  $180^\circ \leq \theta \leq 270^\circ$ , find
- (i)  $\sin \theta$  2
- (ii)  $\operatorname{cosec} \theta$  1
- d) Solve for  $0^\circ \leq \theta \leq 360^\circ$ ,  $\sin 2\theta = \frac{1}{2}$  3
- e) If  $\tan^2 \theta + 2 \sec^2 \theta = 5$ , find the value of  $\sin^2 \theta$  3

Question 4 on page 3.

**Question 4** 15 marks Start this question on a new page.

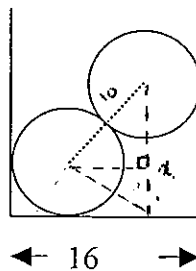
a) Simplify:  $\frac{(\operatorname{cosec} \theta + \cot \theta)(\operatorname{cosec} \theta - \cot \theta)}{(\sec \theta + \tan \theta)(\sec \theta - \tan \theta)}$  3

- b) A soccer goal is 7.3m wide. A Newington player shoots for goal when he is 16m from one goal post and 18m from the other.

(i) Draw a diagram to illustrate the information. 1

(ii) Within what angle must a ground-shot be made to score a goal? (Answer to the nearest minute) 2

- \* c) The diagram shows two identical spheres of radius 5cm in a container. How high is the centre of the higher sphere above the base of the container? Note: The dotted line shown in the diagram joins the centers of the spheres and will also pass through the point of contact of the spheres. 3



- d) Two Newington College Cadet Platoons leave camp at noon in directions  $242^\circ T$  and  $148^\circ T$  at marching rates of 7 and 8 kilometers per hour respectively.

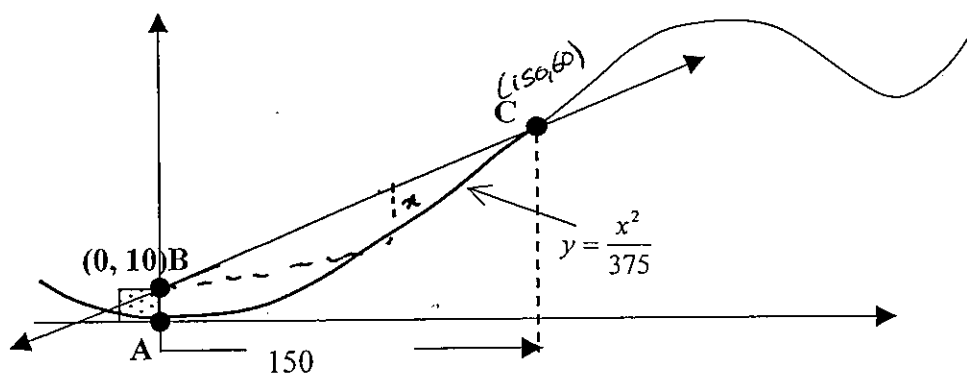
- (i) Draw a diagram showing the information above. 1
- (ii) How far are they apart by 12.45p.m? (Correct to 2 dec.P.). 2
- (iii) What is the bearing of the slower Platoon from the faster Platoon at that time? (Answer to the nearest minute). 3

Question 5 on page 4.

**Question 5** 15 marks Start this question on a new page.

- a) Find the co-ordinates of centre and the length of the radius, of a circle whose equation is:  $x^2 + 6x + y^2 - 4y + 4 = 0$ . 3
- \* b) The line  $x + 2y - 5 = 0$  is at most  $\sqrt{5}$  units from the point  $(k, 3)$ . 3  
What range of values may  $k$  take.
- c) Find the coordinates of the point M which divides the interval joining  $(2, -3)$  and  $(4, 6)$  externally in the ratio 2:1 3
- d) A ski resort plans to open a new slope. The diagram below shows the proposed slope running from the high point, C, down to point A. The straight line BC represents the proposed chair lift. Point B, the start of the chair lift, is to be situated on the roof of an existing building.

The situation can be modeled mathematically if part of the curve  $y = \frac{x^2}{375}$  is used to represent the slope AC, point A has coordinates  $(0, 0)$ , B has coordinates  $(0, 10)$ , C has coordinates  $(150, 60)$  and BC is a straight line.



- (i) Find the length of the straight line BC. 1  
(To the nearest metre.)
- (ii) Find the equation of the straight line BC. 2
- (iii) How high is the chair lift above the snow at the mid point of BC? 2
- (iv) Find the angle of inclination of the chair lift. 1  
(To the nearest minute).

**END OF PAPER**

Question 1 (15 marks)

i)  $f(x) = x^2 + 1 + \frac{1}{x^2}$

$f(y) = y^2 + 1 + \frac{1}{y^2}$

$f\left(\frac{1}{y}\right) = \left(\frac{1}{y}\right)^2 + 1 + \frac{1}{\left(\frac{1}{y}\right)^2}$   
 $= \frac{1}{y^2} + 1 + y^2$

$\therefore f(y) = f\left(\frac{1}{y}\right)$

b)  $y = \sqrt{x-3}$

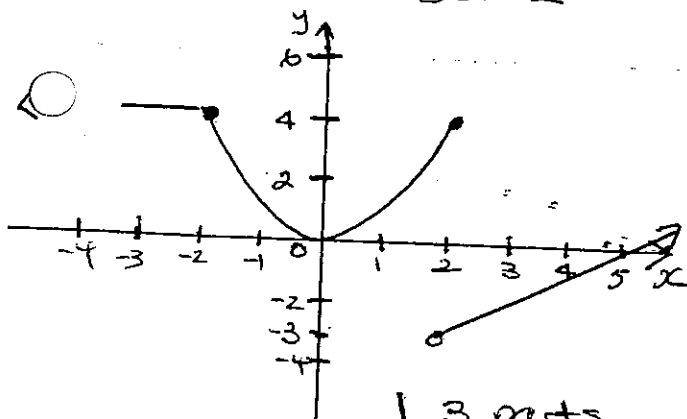
○  $\text{ain: } x-3 \geq 0$   
 $x \geq 3$

$\therefore \text{domain: } x \geq 3$

$\text{range: } y \geq 0$

$\therefore$  i) Sketch

$f(x) = \begin{cases} 4 & x < -2 \\ x^2 & -2 \leq x \leq 2 \\ x-5 & x > 2 \end{cases}$



1 3 parts  
1 details

1  $f(-5) + 3f(-2) - f(2) + 2(f(4))$

$f(-5) = 4$

$f(-2) = 4$

$f(2) = 4$

$f(4) = -1$

$= 4 + 3 \times 4 - 4 + 2 \times -1$   
 $= 10$

(d) i)  $y = \frac{2}{x^2-9}$

$f(x) = \frac{2}{x^2-9}$

$f(x)$  connect  
must be  
there  
somehow

$f(-x) = \frac{2}{(-x)^2-9}$

$f(-x) = \frac{2}{x^2-9}$

$\therefore f(-x) = f(x)$

$\therefore$  it is an even function

(ii) Vertical asymptotes

$y = \frac{2}{x^2-9}$

$x^2-9 \neq 0$

$(x-3)(x+3) \neq 0$

$\therefore x \neq \pm 3$

$\therefore$  vertical asymptotes are

$x=3$  and  $x=-3$

equations

(iii) Horizontal asymptotes

$y = \frac{2}{x^2-9}$

$y = \frac{\frac{2}{x^2}}{\frac{x^2}{x^2} - \frac{9}{x^2}}$

$\lim_{x \rightarrow \infty} y = \frac{0}{1-0}$

$y = 0$

equation.

$\therefore$  horizontal asymptote at  $y=0$

Intercepts

X int  $y=0$

$0 = \frac{2}{x^2-9}$

$0 \neq 2$

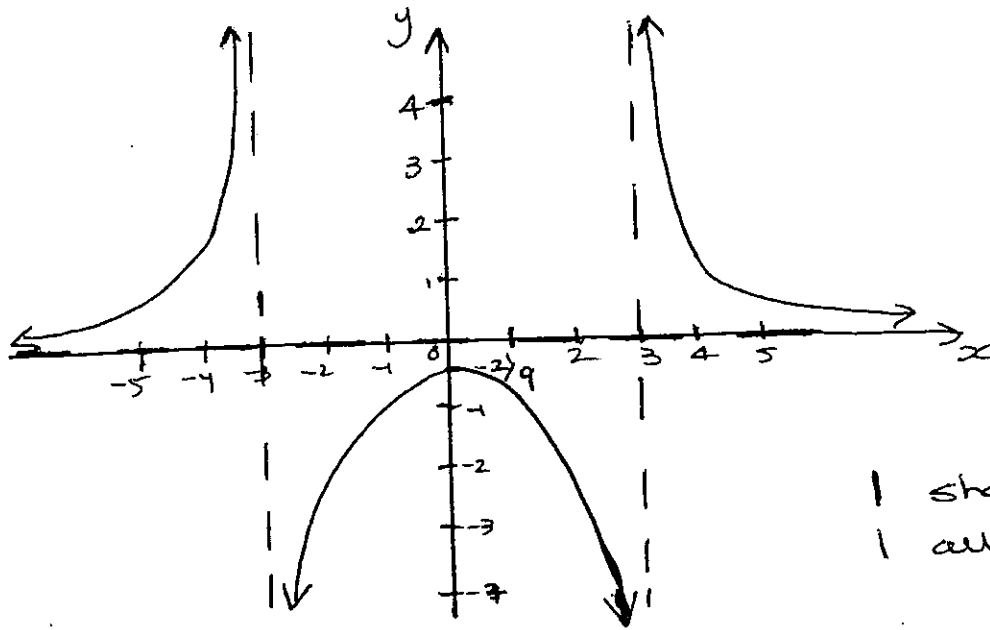
$\therefore$  no x intercepts

Y int  $x=0$

$y = \frac{2}{-9}$

$\left(0, -\frac{2}{9}\right)$

$$y = \frac{2}{x^2 - 9}$$



! shape  
! all information

C

C

$$\frac{2x+3}{5x-1} = \frac{2005}{2x+1}$$

$$(2x+3)(5x-1) = (2x+1)(5x-1)$$

$$10x^2 - 10x + 15x - 3 = 10x^2 - 2x + 5x - 1$$

$$2x = 14$$

$$x = 7$$

(3)

$$x + 2y = -8, \quad xy = 8$$

$$y = \frac{8}{x}$$

$$x + 2\left(\frac{8}{x}\right) = -8$$

$$x^2 + 16 = -8x$$

$$x^2 + 8x + 16 = 0$$

$$(x+4)(x+4) = 0$$

$$x = -4$$

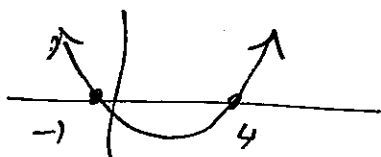
$$\text{Sub } y = \frac{8}{(-4)} = -2$$

$$\text{Ans } \begin{cases} x = -4 \\ y = -2 \end{cases}$$

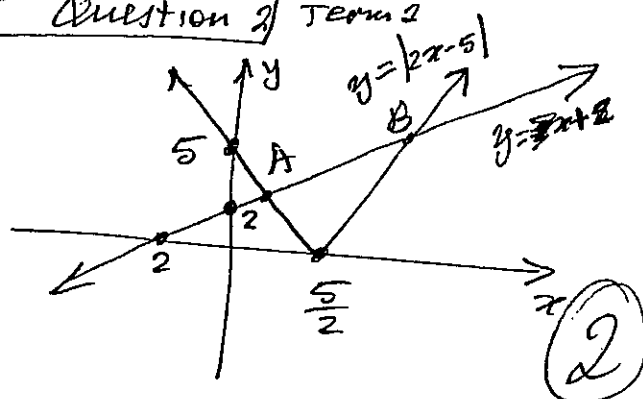
$$4 + 3x - x^2 > 0$$

$$0 > x^2 - 3x - 4$$

$$0 > (x-4)(x+1)$$



$$-1 < x < 4$$



either A

$$-(2x-5) = x+2$$

$$-2x+5 = x+2$$

$$x = 1$$

$$1 \leq x \leq 7$$

OR

B

$$2x-5 = x+2$$

$$x = 7$$

(2)

$$\frac{2}{x-3} < 4$$

$$\frac{2}{(x-3)} (x-3)^2 < 4 (x-3)^2$$

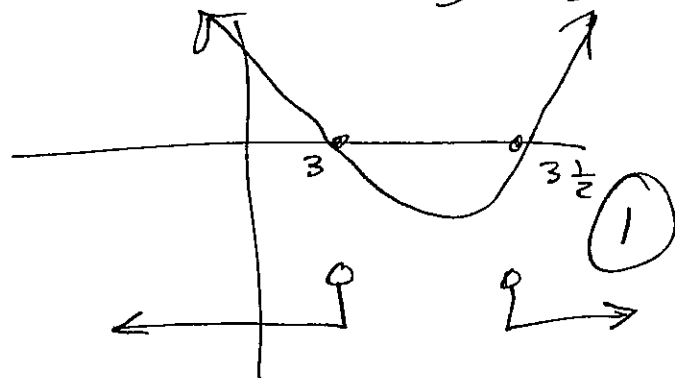
$$2(x-3) < 4(x^2 - 6x + 9)$$

$$2x - 6 < 4x^2 - 24x + 36$$

$$0 < 4x^2 - 26x + 42$$

$$0 < 2(2x^2 - 13x + 21)$$

$$0 < (2x-7)(x-3)$$



$$x < 3 \quad x > 3\frac{1}{2}$$

C

O

# Solutions

Q3

$$(a) \sin 45 \cos 30 + \cos 45 \sin 30 = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \quad 2 \text{ mks}$$

$$= \frac{\sqrt{3}+1}{2\sqrt{2}} \quad (\text{correct exact values})$$

$$= \frac{\sqrt{2}(\sqrt{3}+1)}{4}$$

$$= \frac{\sqrt{6}+\sqrt{2}}{4}$$

1mk.  
(rationality den)

$$(b) 2\sin x + \sqrt{3} = 0$$

$$2\sin x = -\sqrt{3}$$

$$\sin x = -\frac{\sqrt{3}}{2}$$

1mk  
(quadrant)



$$\text{let } \sin \alpha = \frac{\sqrt{3}}{2}$$

$$\therefore \alpha = 60^\circ$$

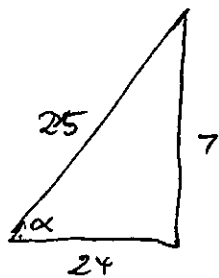
1mk (related angle).

$$\therefore x = 180 + 60 \quad \text{or} \quad 360 - 60$$

$$\therefore x = 240^\circ \quad \text{or} \quad 300^\circ$$

1mk (solution).

$$(c) \tan \theta = \frac{7}{24}, \quad 180^\circ \leq \theta \leq 270^\circ$$



$$\text{let } \theta = 180 + \alpha$$

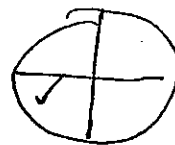
$$\therefore x^2 = 24^2 + 7^2$$

$$\therefore x = 25$$

$$\therefore \sin \alpha = \frac{7}{25}$$

1mk (Pythag).

$$\text{and } \csc \alpha = \frac{25}{7}$$



$$\therefore (i) \sin \theta = \sin(180 + \alpha)$$

$$= -\sin \alpha$$

$$= -\frac{7}{25}$$

1mk (sin)

$$(ii) \csc \theta = -\frac{25}{7}$$

1mk (inverse).

$$(d) \sin 2\theta = \frac{1}{2}; \quad \text{for } 0^\circ \leq \theta \leq 360^\circ$$

$$\therefore 0^\circ \leq 2\theta \leq 720^\circ$$

$$\text{let } \sin \alpha = \frac{1}{2}$$

$$\therefore \alpha = 30^\circ$$

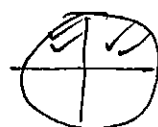
1mk (domain).

1mk (related angle).

$$2\theta = \alpha, 180 - \alpha, 360 + \alpha, 540 - \alpha$$

$$\therefore 2\theta = 30^\circ, 150^\circ, 390^\circ, 510^\circ$$

$$\therefore \theta = 15^\circ, 75^\circ, 195^\circ, 255^\circ$$



$$) \quad \tan^2 \theta + 2 \sec^2 \theta = 5$$

$$\therefore \quad \cancel{\tan^2 \theta} +$$

$$\sec^2 \theta - 1 + 2 \sec^2 \theta = 5$$

(1 mks <sup>sub</sup> identity).

$$\therefore 3 \sec^2 \theta = 6$$

$$\therefore \sec^2 \theta = 2$$

$$\therefore \cos^2 \theta = \frac{1}{2}$$

(1 mks <sup>sub</sup>  $\cos^2 \theta$ ).

$$\text{but } \sin^2 \theta = 1 - \cos^2 \theta$$

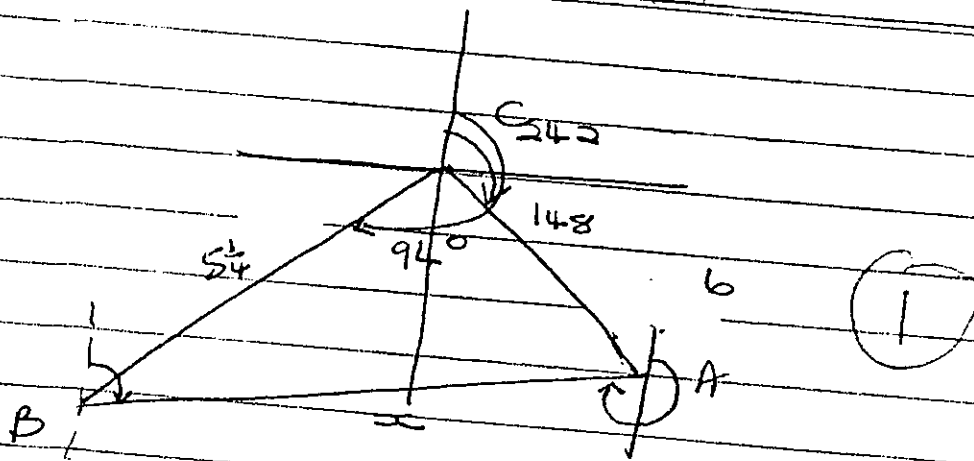
$$\therefore \sin^2 \theta = \frac{1}{2}$$

(1 mks <sup>sub</sup>  $\sin^2 \theta$ ).

C

C

Q4



$$\frac{3}{4} \times 7 = 5 \frac{1}{4}$$

$$\frac{3}{4} \times 8 = 6$$

$$x^2 = 6^2 + \left(5 \frac{1}{4}\right)^2 - 2 \times 6 \times 5 \frac{1}{4} \cos 94^\circ$$

$$= 67.95715785$$

$$x = 8.243613155$$

$$= 8.24 \text{ to 2dp}$$

$$\frac{\sin A}{5 \frac{1}{4}} = \frac{\sin C}{x}$$

$$\frac{\sin A}{5 \frac{1}{4}} = \frac{\sin 94^\circ}{8.24}$$

$$\sin A = 5 \frac{1}{4} \times \sin 94^\circ$$

$$= 0.635583891$$

$$A = 39^\circ 27' 47.89''$$

exact  
0.635305316  
39° 26' 46.26"

$$\text{Bearing} = 360^\circ - 32^\circ - A$$

$$288^\circ 53'$$

(Q4)

$$a) \frac{(\operatorname{cosec} \theta + \cot \theta)(\operatorname{cosec} \theta + \cot \theta)}{(\sec \theta + \tan \theta)(\sec \theta - \tan \theta)}$$

$$= \frac{\operatorname{cosec}^2 \theta - \cot^2 \theta}{\sec^2 \theta - \tan^2 \theta} \quad (1)$$

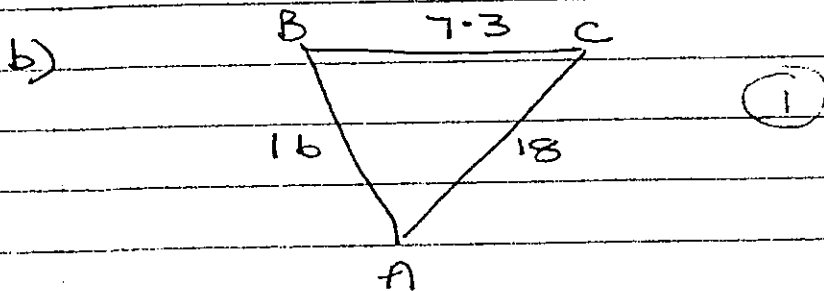
$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \cot^2 \theta = \sec^2 \theta$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$= \frac{1 + \cot^2 \theta - \cot^2 \theta}{\tan^2 \theta + 1 - \tan^2 \theta} \quad (1)$$

$$= 1 \quad (1)$$



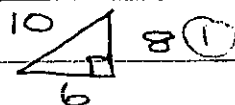
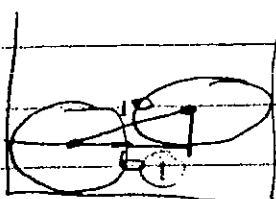
$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$= \frac{18^2 + 16^2 - 7.3^2}{2 \times 18 \times 16} \quad (1)$$

$$= \frac{526.71}{576}$$

$$A = 23^\circ 52' 31.81''$$

$$A = 23^\circ 53' \quad (1) \text{ to nearest min}$$



$$d = 8 + 5$$

$$= 13 \quad (1)$$

# Q5 - Solution [out of 12]

13

a)  $x^2 + 6x + y^2 - 4y + 4 = 0$

$$(x+3)^2 + (y-2)^2 = -4 + 9 + 4 \quad \checkmark$$

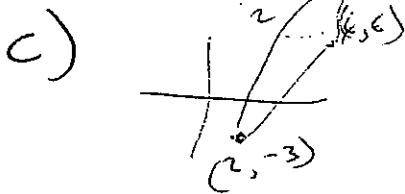
$$(x+3)^2 + (y-2)^2 = 9$$

Centre  $(-3, 2)$   $\checkmark$

radius 3  $\checkmark$

3

b) Delete ~~3 marks~~ for all candidates [Answer ~~is~~  $(-3, 2)$ ]



$$P(x, y) = P\left(\frac{2 \times 4 - 2}{2 - 1}, \frac{2 \times 6 + 3}{2 - 1}\right) \quad \checkmark \checkmark$$

$$= (6, 15) \quad \checkmark \quad \boxed{3}$$

(d) (i)  $BC = \sqrt{(60 - 10)^2 + (150 - 0)^2}$

$$= \sqrt{50^2 + 150^2}$$

$$= \underline{158} \text{ m [correct]} \quad \checkmark \quad \boxed{1}$$

ii)  $m = \frac{60 - 10}{150}$

$$= \frac{1}{3} \quad \checkmark \quad y - 10 = \frac{1}{3}(x - 0)$$

$$3y - 30 = x \quad \checkmark \quad \boxed{2}$$

$$x - 3y + 30 = 0 \neq \left\{ y = \frac{1}{3}x + 10 \right\}$$

(iii) mid pt BC =  $\left(\frac{150}{2}, \frac{20}{2}\right)$

$$= (75, 35) \quad \checkmark \quad \boxed{2} + \checkmark = \boxed{3}$$

$$\therefore y = \frac{75^2}{375} \quad \checkmark$$

$$= \underline{15} \text{ m} \quad \therefore 20 \text{ m above snow.}$$

(iv) Angle of inclination =  $\tan^{-1}\left(\frac{50}{150}\right)$

$$= \underline{18^\circ 26'} \quad \checkmark \quad \boxed{1}$$